

Final Assessment Spring 2020.

Q1. Quadratic time is faster than

- (a) $\Theta(1)$; (b) $\Theta(\lg n)$; (c) $\Theta(n^2)$; (d) $\Theta(n^3)$;

Q2. What is the complexity of left-to-right binary exponentiation algorithm to compute a^n .

- a. $O(n)$
b. $O(n^2)$
c. $O(n \log n)$
d. $O(\log n)$

Q3. What is the exact number of multiplications made by the following algorithm?

ALGORITHM *BruteForcePolynomialEvaluation*($P[0..n], x$)

//Computes the value of polynomial P at a given point x

//by the "highest to lowest term" brute-force algorithm

//Input: An array $P[0..n]$ of the coefficients of a polynomial of degree n ,

// stored from the lowest to the highest and a number x

//Output: The value of the polynomial at the point x

$p \leftarrow 0.0$

for $i \leftarrow n$ **downto** 0 **do**

$power \leftarrow 1$

for $j \leftarrow 1$ **to** i **do**

$power \leftarrow power * x$

$p \leftarrow p + P[i] * power$

return p

- a. $2n+1$
b. $n(n-1)/2$
c. $n(n+1)/2$
d. $(n+1)(n+2)/2$

Q4. What is the complexity of Gaussian elimination?

- a. n
b. n^2
c. n^3
d. $n^{2.87}$

Q5. What is the exact number of comparisons made by the fake coin problem in the worst case when solved by the brute force technique and if we consider that the number of input coins is odd?

- a) n
b) $n/2$

- c) $(n-1)/2$
- d) $(n+1)/2$

Q6. The running time function of an algorithm is determined by

- (a) the number of operations in a sequence structure; (b) the number of branches in a selection structure; (c) the time of the slowest of a series of loops; (d) the data

Q7. After each step of the binary search, the quantity of remaining data to be searched is on average:

- (b) 1; (b) $\lg n$; (c) $n \div 2$; (d) n ; (e) $2n$

Q8. A recurrence defines

- (a) a set of natural numbers; (b) a logical formula; (c) a computable function; (d) an undecidable problem;

Q9. What's the largest number of key comparisons performed by the merge phase of mergesort for an n -element array?

- a) n b) $n + 1$ c) $n - 1$ d) $n / 2$

Q10. A recursive function definition

- (a) uses a *while* loop; (b) lists all possibilities; (c) contains a call to the function itself; (d) is inefficient

Q11. Recurrences are used in

- (a) input specification; (b) proofs of correctness; (c) time analysis; (d) type checking;

Q12. Graph path search involves finding a:

- (a) set of vertices; (b) sequence of vertices; (c) set of edges; (d) minimal set of edges;

Q13. The common implementation of a heap is

- (a) array; (b) linked list; (c) doubly-linked structure; (d) multilinked structure; (e) none of these

Q14. The root node of a binary search tree contains:

- (a) the lowest value in the tree; (b) a value not higher than that stored in its left child; (c) a value not higher than that stored in its right child; (d) the highest value stored in the tree; (e) no value

Q15. The running time for *heap-sort* is:

- (a) $O(1)$; (b) $O(\lg n)$; (c) $O(n \lg n)$; (d) $O(n)$; (e) $O(n^2)$

Q16. The running time for *build-heap* is:

- (a) $O(1)$; (b) $O(\lg n)$; (c) $O(n \lg n)$; (d) $O(n)$; (e) $O(n^2)$

Q17. In a recursive algorithm, when the running time for the base case is $O(1)$ and remaining work of an algorithm is reduced by one at each step, the running time is:
(a) $O(1)$; (b) $O(\lg n)$; (c) $O(n \lg n)$; **(d) $O(n)$** ; (e) $O(n^2)$

Q18. In a recursive algorithm, when the running time for the base case is $O(n)$ and remaining part of input to process is reduced by one at each recursive step, the total running time is:
(a) $O(1)$; (b) $O(\lg n)$; (c) $O(n \lg n)$; (d) $O(n)$; **(e) $O(n^2)$**

Q19. When the quantity of remaining data to be processed in an algorithm, at each step, is $(n \div 2)$, then the total complexity is: (a) $O(1)$; **(b) $O(\lg n)$** ; (c) $O(n)$; (d) $O(n^2)$; (e) $O(2^n)$

Q20. What's the efficiency of the recurrence $A(n) = 2 A(n/2) + 2n, A(1) = 0$?

A) $\Theta(n)$

B) $\Theta(1)$

C) $\Theta(n^2)$

D) $\Theta(n \log n)$

E) $\Theta(n^{\log_2 n})$